



DELHI PUBLIC SCHOOL NEWTOWN

SESSION: 2022-23

HALF YEARLY EXAMINATION

CLASS: IX
SUBJECT: CHEMISTRY [SET B]

FULL MARKS: 80
DATE: 30.08.2022

General Instructions:

- The paper consists of six printed pages.
- Section A is compulsory. Attempt any four questions from Section B.
- Answers should be to the point.
- Question numbers should be copied carefully while answering the questions.

SECTION A

(Attempt all questions from this section)

Question 1

Choose one correct answer to the questions from the given options:

[15]

- i) The molecular formula of Ammonium phosphate is:
- (a) NH_4PO_3
 - (b) $\text{NH}_4(\text{PO}_3)_2$
 - (c) $(\text{NH}_4)_3\text{PO}_4$
 - (d) $(\text{NH}_4)_3(\text{PO}_4)_2$
- ii) A divalent metal which displaces hydrogen from cold water is:
- (a) Sodium
 - (b) Calcium
 - (c) Iron
 - (d) Zinc
- iii) An anion having a metallic element in it is:
- (a) Aluminate
 - (b) Phosphate
 - (c) Borate
 - (d) Chlorite
- iv) The chemical name of HCOOK is:
- (a) Potassium carbonate
 - (b) Potassium formate
 - (c) Potassium acetate
 - (d) Potassium oxalate

- v) The substance used to remove phosphine from hydrogen is:
- Anhydrous calcium chloride
 - Potassium hydroxide solution
 - Silver nitrate solution
 - Lead nitrate solution
- vi) The salt which on reaction with sodium hydroxide gives a dirty green precipitate is:
- Ferric sulphate
 - Copper sulphate
 - Zinc sulphate
 - Ferrous sulphate
- vii) The metal which react with copper sulphate to give a reddish brown residue is:
- Silver
 - Iron
 - Mercury
 - Platinum
- viii) The symbolic representation of sub atomic particles is:
- ${}_1n^0, {}_1p^0, {}_{-1}e^1$
 - ${}_1p^1, {}_1n^0, {}_{-1}e^0$
 - ${}_{-1}e^0, {}_0n^1, {}_1p^{+1}$
 - ${}_{-1}e^0, {}_1p^1, {}_0n^1$
- ix) An atom has 18 neutrons, 17 protons. Select the correct option which fits well for the valency and type of the element respectively
- +1, metal
 - 1, non metal
 - 0, noble gas
 - 2+, metal
- x) An atom ${}_{19}X^{39}$ having unstable electronic configuration forms a compound with ${}_{16}Y^{32}$. The bond formed is ____ and the formula of the compound is ____.
- electrovalent, X_2Y
 - electrovalent, XY_2
 - covalent, X_2Y
 - covalent, XY_3
- xi) At constant pressure, if the temperature is tripled for a fixed mass of a gas then its volume will become:
- 6 times
 - 1/3 times
 - 3 times
 - no change

- xii) Gas that turns lead acetate paper silvery black and it evolved with rotten egg smell is:
- Carbon monoxide
 - Carbon dioxide
 - Hydrogen
 - Hydrogen sulphide
- xiii) When the precipitation takes place between Sodium hydroxide and zinc chloride, the colour of the precipitate formed is:
- whitish grey
 - bluish white
 - dense white
 - gelatinous white
- xiv) Which of the following pairs represent two atoms with the same number of neutrons?
- ${}^9\text{F}^{19}$ and ${}^{10}\text{Ne}^{20}$
 - ${}^6\text{C}^{12}$ and ${}^{12}\text{Mg}^{24}$
 - ${}^{11}\text{Na}^{23}$ and ${}^{19}\text{K}^{39}$
 - ${}^{27}\text{Co}^{59}$ and ${}^{28}\text{Ni}^{59}$
- xv) STP conditions:
- 273K, 270mm Hg
 - 273K, 760mm Hg
 - 0°C, 76mm Hg
 - 273K, 76mm Hg

Question 2

- i) Write the formula for the following compounds: [5]
- Magnesium nitrite
 - Sodium peroxide
 - Potassium chromate
 - Ammonium carbonate
 - Sodium permanganate
- ii) Balance the following equations: [5]
- $\text{Pb}_3\text{O}_4 + \text{HCl} \rightarrow \text{PbCl}_2 + \text{H}_2\text{O} + \text{Cl}_2$
 - $\text{AlN} + \text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 + \text{NH}_3$
 - $\text{CuS} + \text{O}_2 \rightarrow \text{CuO} + \text{SO}_2$
 - $\text{NH}_3 + \text{Cl}_2 \rightarrow \text{N}_2 + \text{NH}_4\text{Cl}$
 - $\text{C}_2\text{H}_6 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
- iii) State one appropriate observation for the following chemical reaction taking place: [5]
- Barium chloride solution is added to an aqueous solution of sodium sulphate.
 - Sulphur dioxide is added to acidified potassium dichromate solution.
 - Ammonium chloride and calcium hydroxide is heated gently in a test tube.

- (d) Lead (II) carbonate is heated in a test tube.
- (e) Ferric sulphate solution is made to react with sodium hydroxide solution.
- iv) Solve the following numericals : [5]
- (a) A certain volume of gas at 27°C , 1atm pressure occupies a volume of 25 m^3 . If the pressure is kept constant and the temperature is raised to 77°C , what would be the volume of the gas
- (b) If at constant temperature 10mL of a gas is at pressure of 1 atm and we double the pressure, what will be the new volume of the gas?
- v) When white sample [M] is strongly heated, it produces a residue [N] which is yellow when hot and white when cold. The heating also produces a neutral gas [O] which rekindles a glowing wooden splinter and a reddish brown acidic gas [P] which has an irritating odour and dissolves in water to produce two acids.
- (a) Identify M, N, O and P.
- (b) Write a balanced chemical equation for the heating of the compound M. [5]

SECTION B

Attempt any four questions

Question 3

- i) Complete the table given below. [5]

Atom	Mass no	Atomic no	No of Neutrons
$_{19}\text{K}^{39}$	i) _____	19	ii) _____
$_5\text{B}^{10}$	10	iii) _____	iv) _____
$_{20}\text{Ca}^{40}$	v) _____	20	20

- ii) Give reasons for the following: [5]
- (a) Inert gases are unreactive.
- (b) Atomic mass of chlorine is fractional and not in whole numbers.
- (c) It is necessary to compare the gases at STP.
- (d) Nitric acid is not used during the laboratory preparation of hydrogen.
- (e) Silver bromide salt is kept in dark coloured bottles.

Question 4

- i) The formula of sulphide of a metal M is M_2S_3 . Write down the formula of its : [4]
- (a) Bisulphite
- (b) Oxalate
- (c) chlorate
- (d) bromide

- ii) Electronic configuration of an atom X is (2, 8, 7). It combines with another atom Y having atomic number 12. [4]
- Write the electronic configuration of Y
 - Write the formula of the compound formed.
 - What type of bond will be formed between X and Y?
 - Draw the electron dot cross diagram to represent the formation of the above compound.
- iii) Name the law that studies the relationship between the absolute temperature (T) and volume (V) of a fixed mass of a dry gas at constant pressure. Also represent it graphically (V vs T). [2]

Question 5

- i) A gas at constant temperature is at a pressure of 1080mm Hg. If the volume is decreased by 40 %, [5]
- Find its volume at 127°C.
 - State the law that obeys the above statement.
 - Give its mathematical expression.
- ii) Elements A, B and C have atomic numbers 10, 11, 17 respectively. [5]
- Identify a metal, non metal and inert gas among these.
 - Write the formula of the compound formed in between the metal and non metal.
 - Draw the orbit diagram for the compound formed above.

Question 6

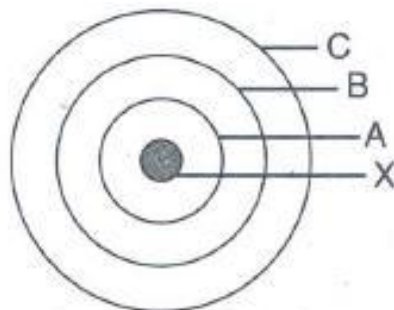
- i) With respect to the industrial preparation of hydrogen answer the following: [5]
- Write the chemical equation for the reaction where heat is absorbed.
 - Name the catalyst and promoter used in this process.
 - Write an equation for the catalysed reaction taking place.
 - How is unused carbon dioxide removed from the hydrogen formed?
- ii) A hydrate of magnesium sulphate $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ contains water of crystallization. Calculate the mass percent of water of crystallisation. [Mg=24, S=32, O=16, H=1] [2]
- iii) Complete the oxidation- reduction reactions: [3]
- $\text{Pb}^{2+} \rightarrow \text{Pb}^{4+}$
 - $\text{Hg}^{2+} \rightarrow \text{Hg}$
 - $\text{F}^- \rightarrow \text{F}$

Question 7

- i) Calculate the volume of ammonia gas at STP which at 17°C and pressure of 740mm of mercury occupies a volume of 6.4cc. [3]
- ii) Write balanced equations for the following conversions: [3]
- (a) Copper (II) nitrate to Copper (II) oxide
 - (b) Zinc to potassium zincate
 - (c) Magnesium to magnesium sulphate
- iii) Differentiate between the following terms: [4]
- (a) Endothermic and exothermic reaction(definition)
 - (b) Neutralization and precipitation reaction(Equation)
 - (c) Atomic number and mass number(definition)
 - (d) Oxidation and reduction (in term of loss or gain of electrons)

Question 8

- i) In the given figure: [5]
- (a) Which shell has maximum energy?
 - (b) What are the particles present in X?
 - (c) State the charge of X.
 - (d) Maximum number of electrons B can have.
 - (e) If C has 2 electrons, state whether it forms a cation or an anion.



- ii) A small piece of iron metal is put into a small trough containing steam. There is an effervescence of gas. [5]
- (a) Name the gas formed in the reaction.
 - (b) Give one chemical test for the identification of the gas.
 - (c) Write an equation for the reaction taking place.
 - (d) State the type of reaction taking place.
 - (e) What will be the compound formed when this gas reacts with nitrogen?